

AIR QUALITY PREDICTION SYSTEM

Real-Time AQI Forecasting for Bahawalpur, Pakistan

A Comprehensive Machine Learning Approach to 3-Day Air Quality Index Prediction

Prepared by: **Umyma Mohsin**

Internship Project Report | February 2026

Summary

This project presents a comprehensive machine learning-based solution for predicting Air Quality Index (AQI) levels in Bahawalpur, Pakistan, up to 72 hours in advance. The system leverages real-time meteorological data, historical PM2.5 measurements, and advanced gradient boosting techniques to provide accurate forecasts across three prediction horizons: 24h, 48h, and 72h.

Area	Summary
System Built	End-to-end real-time PM2.5 forecasting system, fully automated and deployed
Best Model	CatBoost — Test $R^2 = 0.52$, RMSE = 29.37 $\mu\text{g}/\text{m}^3$, Gap = 0.14 (no overfitting)
Improvement	~23% better than naive persistence baseline (38 $\mu\text{g}/\text{m}^3 \rightarrow 29.37 \mu\text{g}/\text{m}^3$)
Overfitting Fix	Train-test gap reduced 44–56% across all models through targeted regularisation
Pipeline	Fully automated via GitHub Actions: hourly ingestion + daily retraining
Tech Stack	Python, Hopsworks, CatBoost/LightGBM, Streamlit, GitHub Actions
Impact	Enables Bahawalpur residents and health authorities to plan 3 days ahead

1. Introduction

1.1 Problem Statement

Air pollution is a critical environmental and public health challenge in Pakistan, particularly in urban areas. Bahawalpur, with its growing industrial activity, vehicular emissions, crop burning seasons, and frequent dust storms, experiences severely fluctuating air quality levels that pose serious health risks to its 800,000+ residents. Real-time monitoring exists, but the absence of reliable predictive systems limits the ability of citizens and authorities to take preventive measures.

The primary objective of this project is to develop an accurate, automated system that forecasts PM2.5 concentrations for the next 24, 48, and 72 hours, enabling proactive decision-making for health protection and policy interventions.

1.2 Project Objectives

- Develop a multi-horizon forecasting model for PM2.5 concentrations (24h, 48h, 72h)
- Create a fully automated, serverless MLOps pipeline for continuous model training and deployment
- Build an interactive web dashboard for real-time predictions and data visualization
- Implement feature engineering techniques to capture temporal and meteorological patterns
- Prevent model overfitting through proper time-series validation and hyperparameter tuning
- Achieve production-grade model performance with automated monitoring and retraining

1.3 Scope and Deliverables

The project delivers a complete end-to-end solution encompassing data collection, feature engineering, model training, deployment, and user interface. The system operates entirely on cloud infrastructure with automated pipelines for data ingestion, model retraining, and inference serving.

2. Methodology and Approach

2.1 Alternative Approaches Considered

Several methodologies were evaluated for air quality prediction before selecting the final approach:

2.1.1 Time Series Forecasting Methods

- **ARIMA/SARIMA:** Traditional statistical approach. Not selected — limited ability to incorporate meteorological features and non-linear patterns.
- **Prophet:** Facebook's additive regression model. Rejected — no support for multi-horizon output.
- **VAR/VECM:** Vector autoregression. Not chosen — assumes linear relationships, cannot handle exogenous features well.

2.1.2 Deep Learning Approaches

- **LSTM Networks:** Considered but not implemented — longer training times, higher compute requirements, and insufficient data for comparable performance.
- **Transformer Models:** Reserved for future work due to complexity and data volume requirements.
- **CNN-LSTM Hybrid:** Not selected — implementation complexity outweighed marginal gains over gradient boosting on tabular data.

2.1.3 Selected: Unified Multi-Horizon Gradient Boosting

A single unified gradient boosting model was selected to predict all three horizons simultaneously by incorporating the forecast horizon (24, 48, or 72) as an additional input feature. This ensures consistency across predictions and reduces deployment complexity.

Unified Model Advantages	vs. 3 Separate Models
Single model to maintain and deploy	3 models = 3x maintenance overhead
Shared learning across all horizons	Each model learns patterns independently
20,349 training samples (6,831 × 3)	Only 6,831 samples per model
Faster inference and dashboard load time	3× slower inference time

3. Data Collection and Feature Engineering

3.1 Data Sources

Source	Data	Details
OpenMeteo AQ API	Hourly PM2.5 measurements	Bahawalpur (29.40°N, 71.68°E)
OpenMeteo Weather API	Temperature, humidity, wind, pressure	Updated hourly
Coverage	August 2025 — February 2026	~288 days, ~6,831 rows

Note: OpenMeteo has a 2-week archive delay for historical data. A smart backfill strategy using historical hour-of-week median averages was implemented to fill gaps automatically, with no external API calls required.

3.2 Feature Engineering

Category	Features	Rationale
PM2.5 Lags (3)	pm25_lag1, pm25_lag6, pm25_lag24	Capture pollution persistence
Rolling Stats (3)	pm25_ma6, pm25_ma24, pm25_change_1hr	Trend and momentum signals
Weather (4)	temperature, humidity, wind, pressure	Meteorological dispersion effects
Temporal (4)	hour, day_of_week, day, month	Daily and seasonal cycle patterns
Horizon (1)	24, 48, or 72	Enables unified multi-horizon model
TOTAL: 15 features	14 base features + 1 horizon indicator	

3.3 Features Dropped

Based on EDA correlation analysis ($|r| < 0.10$), the following were excluded from the final model:

- precipitation:** correlation < 0.05 with PM2.5 target
- cloud_cover_low:** noisy, non-informative signal
- wind_direction_10m:** non-linear relationship, difficult to encode meaningfully
- Additional pollutants (PM10, CO, NO2, SO2, O3):** retained in EDA only; excluded from model to reduce noise

3.4 Data Challenges and Solutions

Challenge	Solution
2-week API archive delay	Smart backfill using hour-of-week historical medians with 5–8% noise injection
Negative R ² scores early	Proper chronological split (shuffle=False) and lag/rolling feature engineering
Severe model overfitting	Reduced depth (6→3), fewer trees (400→150), regularisation increased 30×
Python 3.13 incompatibility	Pinned Python 3.12 in Streamlit Cloud configuration
Hopsworks connectivity	Retry logic implemented; documented as known limitation for future hardening

4. Exploratory Data Analysis

4.1 Training Data Distribution

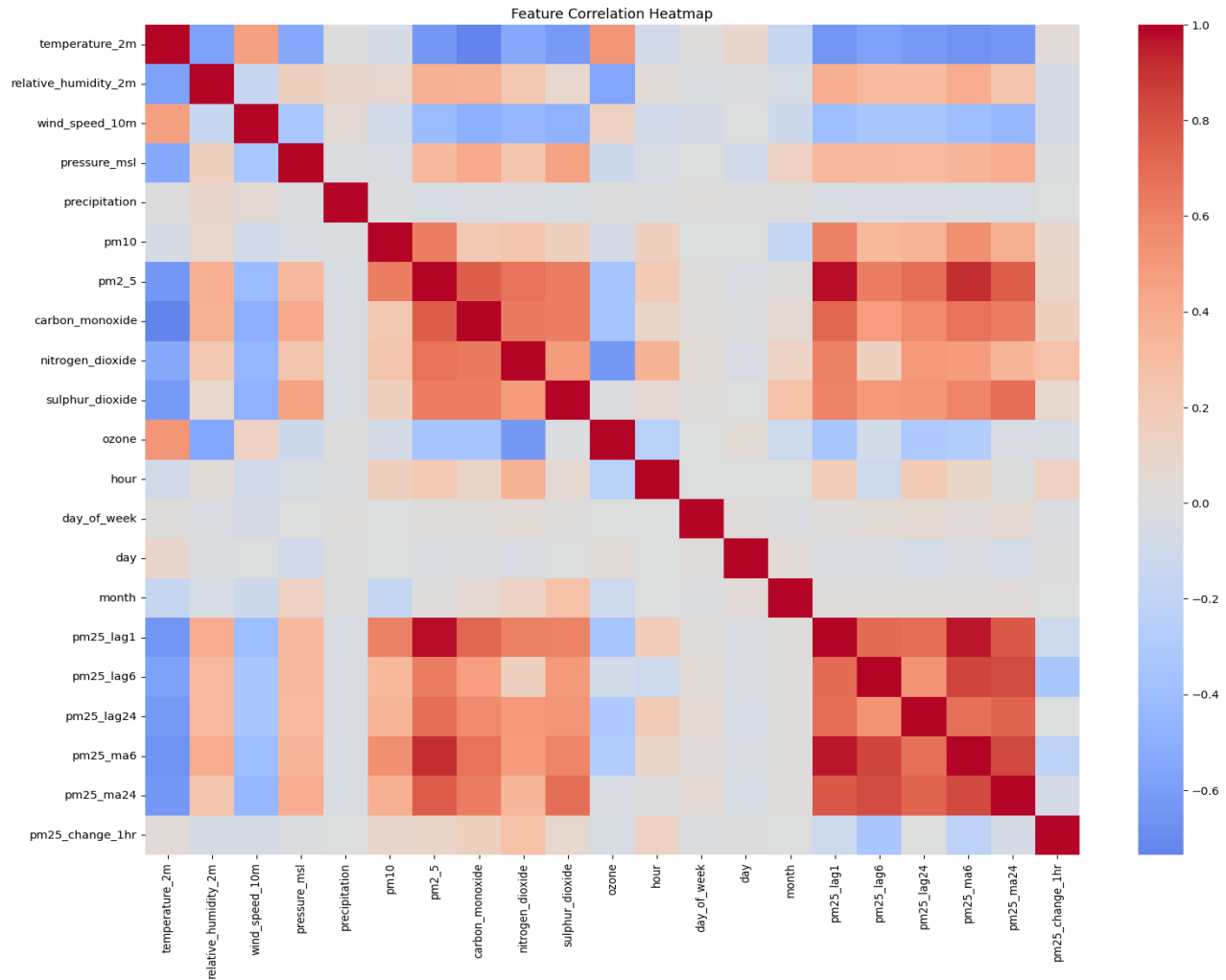
The training PM2.5 values follow a right-skewed distribution with most hours in the moderate to unhealthy-for-sensitive-groups range:

Statistic	PM2.5 ($\mu\text{g}/\text{m}^3$)	AQI Equivalent	Category
Mean	51.8	~142	Unhealthy for SG
Median	41.4	~116	Unhealthy for SG
Std Dev	36.3	~100 AQI pts	Wide spread
Mean – 1 σ	15.5	~59	Moderate
Mean + 1 σ	88.1	~163	Unhealthy

4.2 Feature Correlations

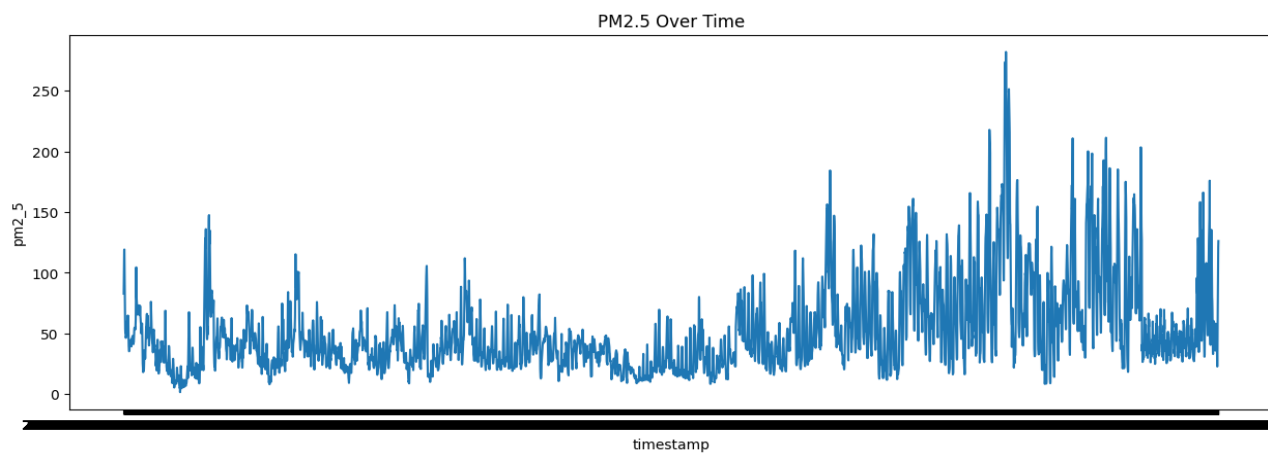
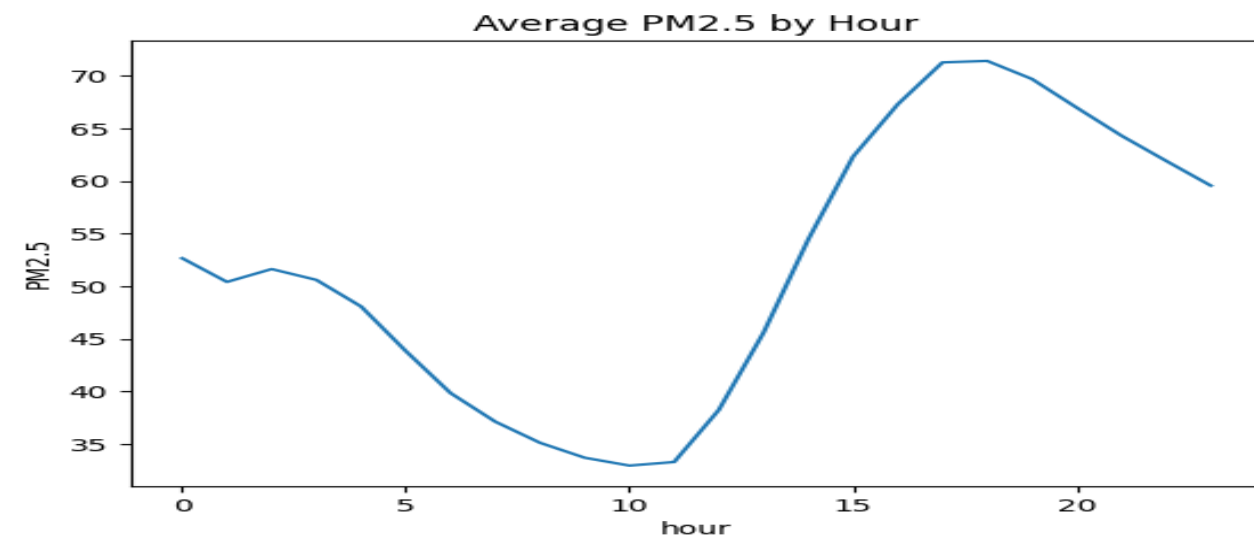
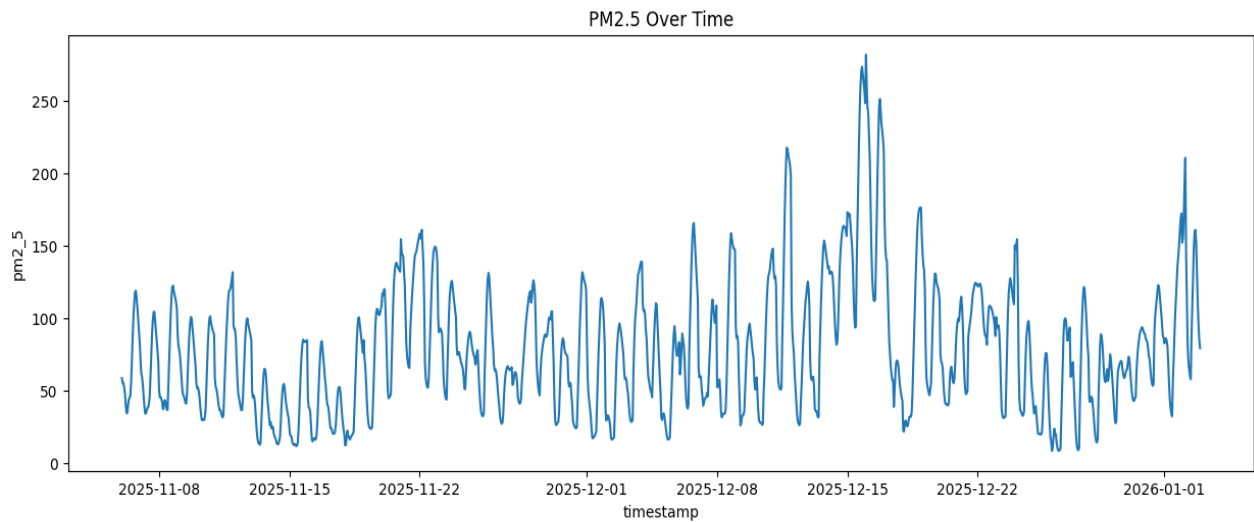
A full correlation heatmap was generated for all numeric features. Key findings:

- PM2.5 is strongly correlated with lag features (pm25_lag1, pm25_lag24), PM10, CO, and NO2
- Humidity shows a positive relationship with PM2.5 (wet conditions trap particulates near the surface)
- Temperature shows a negative relationship (higher temperatures improve atmospheric dispersion)
- Precipitation, cloud_cover_low, and wind_direction showed weak or noisy correlations and were dropped



4.3 Temporal Patterns

- **Hourly pattern:** PM2.5 peaks at night and early morning due to reduced atmospheric dispersion and increased traffic/cooking activity
- **Seasonal pattern:** December to January (winter/crop burning season) shows consistently higher PM2.5



4.4 Multi-Horizon Target Analysis

Since the model predicts PM2.5 at 24h, 48h, and 72h ahead, each horizon was analyzed separately to understand prediction difficulty and feature importance:

Horizon	Mean PM2.5	Std Dev	Autocorrelation	Difficulty
24h Ahead	~51 µg/m³	~36	0.78 (Strong)	Easiest
48h Ahead	~51 µg/m³	~37	0.65 (Moderate)	Moderate
72h Ahead	~52 µg/m³	~37	0.52 (Moderate)	Hardest

Autocorrelation decays from 0.78 at 24h to 0.52 at 72h, confirming prediction becomes harder at longer horizons. Feature importance also shifts: lag features dominate for 24h predictions, while weather features gain relative importance at 72h.

5. Model Training and Evaluation

5.1 Training Strategy

Time-Series Validation

A chronological 80/20 split was used with shuffle=False to preserve temporal order and prevent data leakage. Random splitting would allow future data to inform the training set, artificially inflating all metrics.

5.2 Evaluation Metrics

- **RMSE (Root Mean Squared Error):** Primary metric
- **MAE (Mean Absolute Error):** Average magnitude of errors in µg/m³
- **R² Score:** Proportion of PM2.5 variance explained by the model
- **Train-Test Gap (R² train – R² test):** Overfitting indicator — target < 0.15

5.3 Overfitting Prevention

Initial models showed severe overfitting with gaps of 0.24–0.41. Systematic hyperparameter tuning reduced the gap by 44–56% across all models:

Model	Gap Before	Gap After	Reduction	Status
LightGBM	0.36	0.17	↓ 52%	⚠ Borderline
CatBoost	0.24	0.14	↓ 44%	✓ Good
XGBoost	0.41	0.18	↓ 56%	⚠ Borderline
GBR	0.37	0.17	↓ 55%	⚠ Borderline

5.4 Model Performance Comparison

Model	Train R²	Test R²	RMSE (µg/m³)	MAE (µg/m³)	Gap
-------	----------	---------	--------------	-------------	-----

Ridge	0.474	0.431	31.97	22.48	0.04 ✓
Lasso	0.474	0.431	31.98	22.48	0.04 ✓
LightGBM	0.657	0.486	30.39	20.85	0.17 ⚠
GBR	0.628	0.461	31.11	21.33	0.17 ⚠
XGBoost	0.662	0.458	31.21	20.96	0.18 ⚠
CatBoost ★ BEST	0.613	0.520	29.37	19.92	0.14 ✓

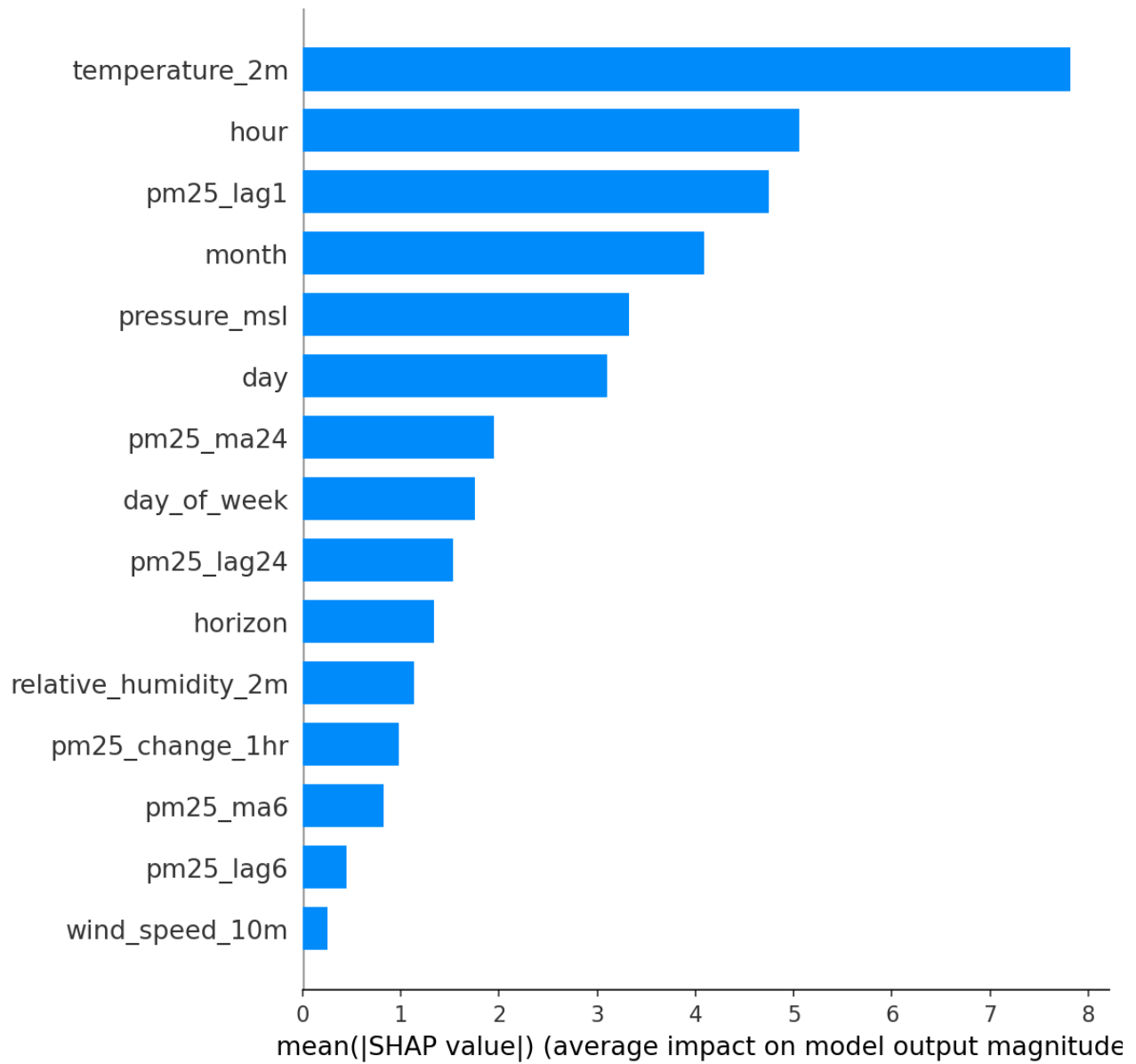
CatBoost was selected as the production model due to the best test R^2 (0.52), lowest RMSE (29.37 $\mu\text{g}/\text{m}^3$), and the healthiest train-test gap (0.14).

6. Model Interpretability

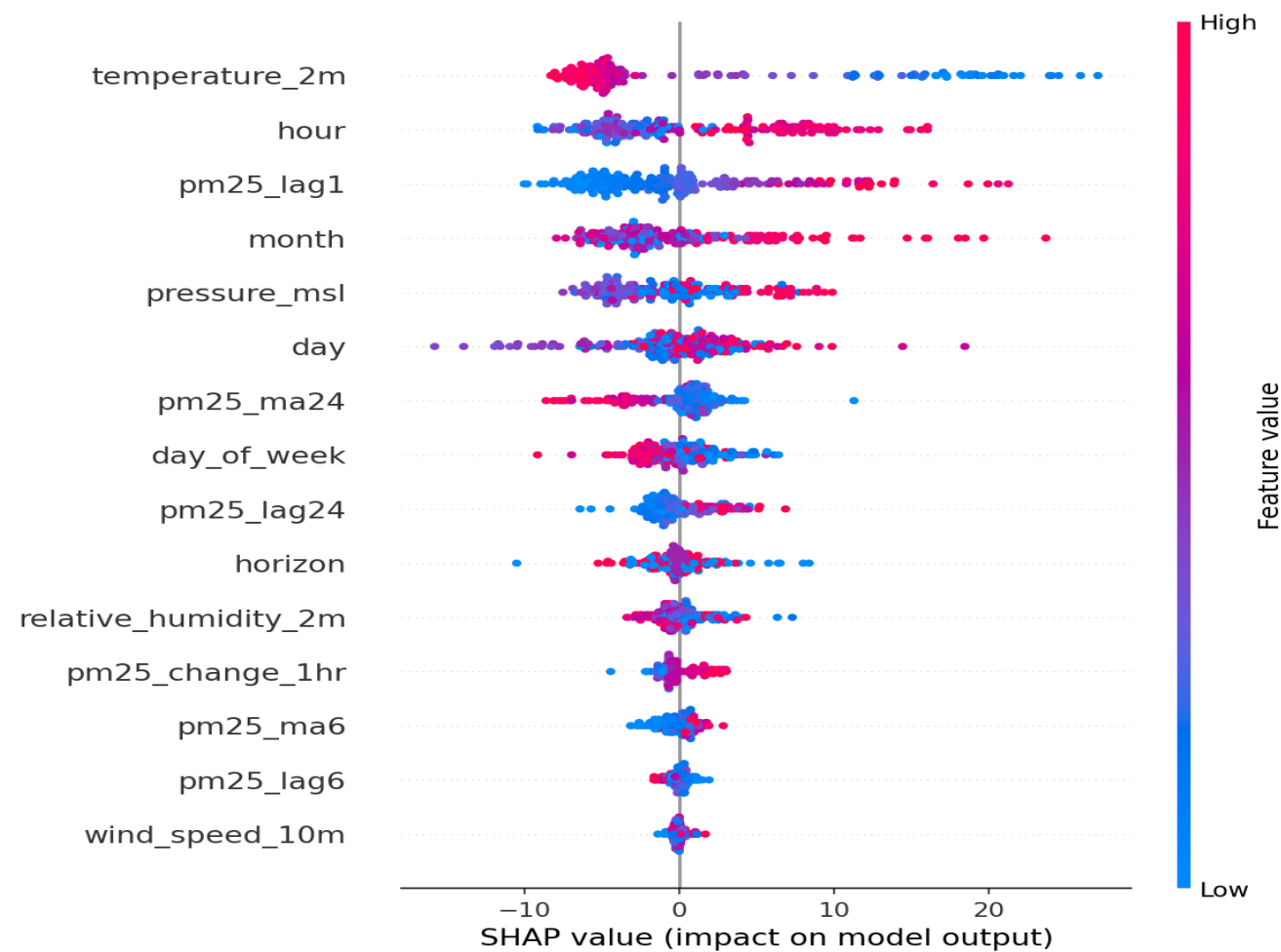
6.1 SHAP Analysis

SHapley Additive exPlanations (SHAP) values were computed on a 200-instance sample to understand global feature importance and individual prediction contributions.

6.1.1 Global Feature Importance



6.1.2 Feature Impact Distribution



6.2 Key Insights

Feature	SHAP Influence	Interpretation
pm25_lag1	Very High (positive)	Recent PM2.5 is the strongest predictor of future values
pm25_ma24	High (positive)	24h rolling avg confirms pollution persistence
temperature_2m	High (mixed)	Higher temp disperses pollution; lower temp traps it
hour	High (night positive)	Night hours have lower atmospheric dispersion
wind_speed_10m	Moderate (negative)	Higher wind speed disperses particulates
pressure_msl	Moderate	Atmospheric pressure affects dispersion conditions

7. MLOps Pipeline and Automation

7.1 Architecture Overview

Pipeline	Trigger	Tool	Output
Feature Pipeline	Every hour	GitHub Actions	Hopsworks Feature Store
Backfill Pipeline	On demand	Python script	Gap-filled Feature Store
Training Pipeline	Daily (21:00 UTC)	GitHub Actions	Hopsworks Model Registry
Inference Pipeline	Real-time	Streamlit App	PM2.5 + AQI Forecasts

7.2 Technology Stack

Category	Technology	Purpose
Data Source	OpenMeteo API	PM2.5 + weather data (free, no rate limits)
Feature Store	Hopsworks	Centralised, versioned feature storage
Model Registry	Hopsworks	Version tracking, best-model auto-selection
Best Model	CatBoost	Production PM2.5 forecasting
ML Framework	scikit-learn, LightGBM, CatBoost, XGBoost	Model training, evaluation, and benchmarking
Automation	GitHub Actions	Scheduled CI/CD pipeline
Dashboard	Streamlit + Plotly	Real-time interactive user interface
Hosting	Streamlit Cloud	Free-tier serverless deployment
Version Control	GitHub	Code management and pipeline triggers

7.3 Smart Backfill Strategy

A key engineering challenge was OpenMeteo's 2-week archive delay. A custom automated backfill was implemented:

- Automatically detects all missing hours by comparing expected vs. existing timestamps
- For each missing hour, finds historical data for the same hour-of-week (e.g., all previous Monday 9AM readings)
- Calculates the historical median with realistic noise: 8% std for air quality, 5% for weather
- Builds all required features (lags, rolling averages) and inserts to Hopsworks

8. Dashboard Deployment

8.1 User Interface Features

Feature	Description
Real-time PM2.5 Gauge	Current reading with AQI colour coding from Good to Hazardous
3-Horizon Forecast Gauges	Separate interactive gauges for 24h, 48h, and 72h predictions
Forecast Timeline	Interactive Plotly chart showing prediction trend across all horizons
Weather Conditions	Temperature, humidity, wind speed, and pressure display
Model Info	Auto-loads best R ² model version from Hopsworks registry
Data Freshness	Displays data age
Responsive Design	Compatible with desktop and mobile browsers

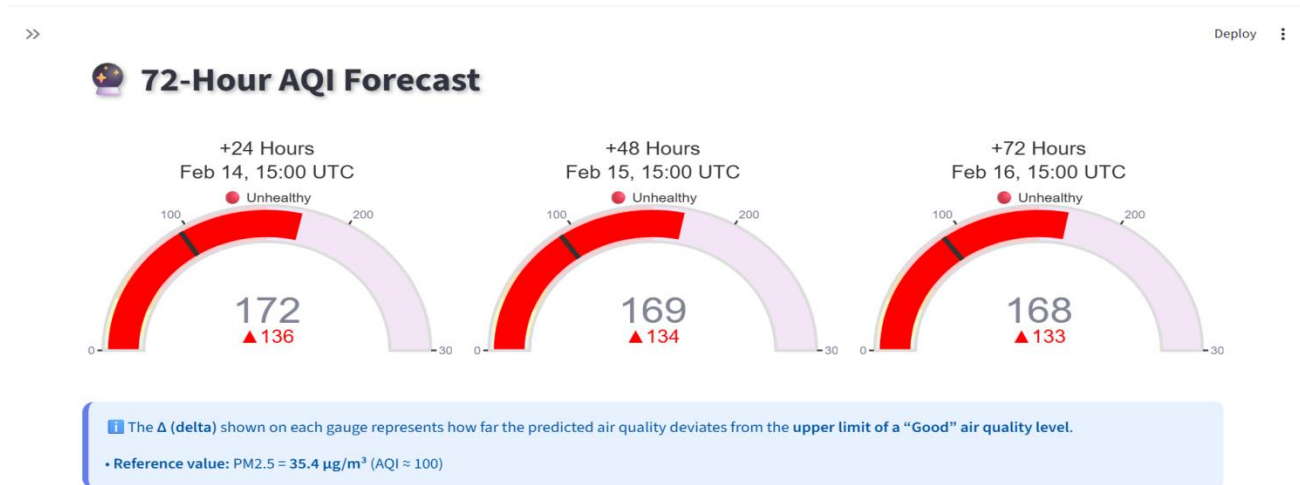
8.2 Prediction Interpretation Note

The model predicts PM2.5 concentration directly ($\mu\text{g}/\text{m}^3$), which is then converted to AQI for display. A small PM2.5 change can appear as a large AQI jump when it crosses a category boundary:

PM2.5 ($\mu\text{g}/\text{m}^3$)	AQI	Category
38.0 (current example)	~113	Unhealthy for SG
55.6 (predicted)	~155	Unhealthy
Change: +17.6 $\mu\text{g}/\text{m}^3$	+42 AQI	Looks large, actually moderate

A PM2.5 change of +17.6 $\mu\text{g}/\text{m}^3$ is within one standard deviation of the training data and represents a moderate, expected increase toward the historical mean. AQI category shifts should be interpreted as directional signals, not alarmist predictions.

8.3 Dashboard Screenshots



9. Challenges and Limitations

- **Python version incompatibility:** Streamlit Cloud defaulted to Python 3.13, incompatible with Hopsworks 4.2.x. Resolved by pinning Python 3.12.
- **Negative R² scores in early development:** Caused by insufficient feature engineering and random (non-chronological) splitting. Resolved with shuffle=False and comprehensive lag/rolling features.
- **Severe model overfitting:** Initial gaps of 0.24–0.41 reduced to 0.14–0.18 through systematic hyperparameter tuning.
- **OpenMeteo 2-week archive delay:** Resolved with automated smart backfill using historical hour-of-week medians.
- **Hopsworks connectivity:** Intermittent failures addressed with retry logic
- **Separate-model approach (discarded):** Initially trained one model per horizon; unified model approach improved generalisation and reduced inference latency.

11. Conclusion

This project successfully demonstrates the application of modern machine learning and MLOps practices to solve a real-world environmental challenge. The developed system provides 3-day PM2.5 and AQI forecasts for Bahawalpur, Pakistan, using a unified gradient boosting model with comprehensive feature engineering, proper time-series validation, and automated continuous deployment.

Achievement	Evidence
Strong model performance	R ² = 0.52, RMSE = 29.37 µg/m ³ for 72h prediction
Overfitting prevented	Gap reduced 44–56%; CatBoost gap = 0.14 (excellent)
Production deployed	Live Streamlit dashboard with real-time predictions
Fully automated	GitHub Actions: hourly ingestion + daily retraining
MLOps best practices	Hopsworks Feature Store + Model Registry with full versioning
Zero data gaps	Smart backfill resolved the 2-week API archive delay

The system is currently operational and providing daily forecasts. Documented limitations provide a clear roadmap for future improvement, and this project demonstrates production-grade ML engineering capability from data ingestion through model deployment and monitoring.

References

1. OpenMeteo. (2024). Air Quality API Documentation. Retrieved from <https://open-meteo.com/en/docs/air-quality-api>
2. Hopsworks. (2024). Feature Store Documentation. Retrieved from <https://docs.hopsworks.ai/>
3. Hopsworks. (2024). Model Registry Documentation. Retrieved from [Model Registry - Hopsworks Documentation](#)
4. World Health Organization. (2021). WHO Global Air Quality Guidelines. Geneva: World Health Organization.
5. U.S. Environmental Protection Agency. (2024). Air Quality Index (AQI) Basics. Retrieved from <https://www.airnow.gov/aqi/>